

Sharp Fourier restriction exponent for Mandelbrot cascades

The squared-critical and squared-supercritical phases

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Status. Candidate partial result, likely valid, pending expert review. The result determines the supremal restriction exponent in the squared-critical and squared-supercritical regimes. It does not decide endpoint attainment or the remaining squared-subcritical regime.

1 Source question

The source is Xinxin Chen, Yong Han, Yanqi Qiu, and Zipeng Wang, *Harmonic analysis of Mandelbrot cascades – in the context of vector-valued martingales*, arXiv:2409.13164v3 [1]. After Corollary 1.6, on page 9, the authors ask:

What is the optimal exponent $r \in (1, 2)$ in

$$\|\widehat{f}\|_{L^2(\mu_\infty)} \leq C(r, \mu_\infty) \|f\|_{L^r(\mathbb{R})}?$$

[Moc00, Theorem 4.1] imply

Corollary 1.6. Assume that $\mathbb{E}[W \log W] < \log b$ and $\mathbb{E}[W^t] < \infty$ for all $t > 0$. Then almost surely on $\{\mu_\infty \neq 0\}$, the measure μ_∞ satisfies the following Fourier restriction estimate: for any $1 \leq r < \frac{4}{4-D_F}$, there exists $C(r, \mu_\infty) > 0$ such that for all $f \in L^r(\mathbb{R})$,

$$(1.12) \quad \|\widehat{f}\|_{L^2(\mu_\infty)} \leq C(r, \mu_\infty) \|f\|_{L^r(\mathbb{R})}.$$

Remark. In obtaining Corollary 1.6, we only use the Fourier dimension and [Moc00, Theorem 4.1]. A natural question is: what is the optimal exponent $r \in (1, 2)$ in the estimate (1.12)?

Figure 1: Corollary 1.6 and the open question in arXiv:2409.13164v3, page 9.

2 Definitions and assumptions

Fix an integer $b \geq 2$. Let $W \geq 0$ be nonconstant with $\mathbb{E}W = 1$, and let μ_∞ be the associated canonical b -adic Mandelbrot cascade on $[0, 1]$. We use the source assumptions

$$\mathbb{E}[W \log W] < \log b, \quad \mathbb{E}[W^t] < \infty \quad (t > 0),$$

and, in the squared-supercritical case, the source's nonlattice assumption on $\log W$. All almost-sure statements below are on $\{\mu_\infty \neq 0\}$.

Set

$$W^{(2)} = \frac{W^2}{\mathbb{E}W^2}.$$

The squared-critical and squared-supercritical regimes are characterized by

$$\mathbb{E}[W^{(2)} \log W^{(2)}] \geq \log b.$$

For a finite Borel measure μ on $\mathbb{R}$, define its Frostman dimension by

$$\dim_{\text{Fr}} \mu = \sup \{ \alpha \geq 0 : \mu(I) \leq C_\alpha |I|^\alpha \text{ for every interval } I \}.$$

For $1 \leq r < 2$, let $\mathcal{R}_r(\mu)$ denote the restriction estimate

$$\|\widehat{f}\|_{L^2(\mu)} \leq C_r \|f\|_{L^r(\mathbb{R})} \quad (f \in \mathcal{S}(\mathbb{R})).$$

Define the supremal restriction exponent

$$r_*(\mu) = \sup\{r \in [1, 2) : \mathcal{R}_r(\mu) \text{ holds}\}.$$

3 Idea of the proof

The source already proves the lower half of the answer: $\mathcal{R}_r(\mu_\infty)$ holds for $r < 4/(4 - D_F)$. The missing sharpness follows from local mass. A restriction estimate at exponent r forces every interval of length δ to have mass $O(\delta^{2/r'})$. Thus restriction cannot pass the exact Frostman dimension.

For Mandelbrot cascades, the Frostman exponent is the maximum of a simple moment profile. The squared-critical condition says that this profile has already stopped increasing by $q = 2$. Rewriting the source's squared-supercritical formula then shows that the Fourier dimension is exactly twice the Frostman dimension. The localization obstruction and the source's positive estimate consequently meet at the same exponent.

4 The partial result

Theorem 1 (Sharp supremal exponent). *Under the assumptions above, suppose*

$$\mathbb{E}[W^{(2)} \log W^{(2)}] \geq \log b.$$

Let $D_F = \dim_F(\mu_\infty)$, whose deterministic value is given by Theorem 1.1 of [1]. Then almost surely on nonextinction,

$$r_*(\mu_\infty) = \frac{4}{4 - D_F}.$$

More precisely, $\mathcal{R}_r(\mu_\infty)$ holds for every $1 \leq r < 4/(4 - D_F)$, and it fails for every $r > 4/(4 - D_F)$.

Proof. Write

$$L(q) = \log \mathbb{E}W^q, \quad a(q) = \frac{q - 1 - \log_b \mathbb{E}W^q}{q} \quad (q > 1).$$

The classical multifractal formula for a nondegenerate Mandelbrot cascade gives, almost surely on nonextinction,

$$\dim_{\text{Fr}} \mu_\infty = \sup_{q>1} a(q). \tag{1}$$

This is the extremal local-dimension, or uniform Holder, part of the standard cascade spectrum; see Molchan [2] and the sharp modulus results of Barral–Kupiainen–Nikula–Saksman–Webb [3].

We first locate the maximum in (1). Direct differentiation gives

$$a'(q) = \frac{\log b + L(q) - qL'(q)}{q^2 \log b}.$$

The numerator

$$H(q) = \log b + L(q) - qL'(q)$$

is nonincreasing, since $H'(q) = -qL''(q) \leq 0$. Nondegeneracy gives

$$H(1) = \log b - \mathbb{E}[W \log W] > 0.$$

On the other hand,

$$\mathbb{E}[W^{(2)} \log W^{(2)}] = 2L'(2) - L(2) \geq \log b$$

is exactly $H(2) \leq 0$. Hence a attains its global maximum at some $q_* \in (1, 2]$, and therefore

$$\sup_{q>1} a(q) = \sup_{1<q\leq 2} a(q). \quad (2)$$

We next compare this maximum with the source formula for D_F . In the squared-supercritical case, formula (1.4) of [1] is

$$D_F = 1 - \inf_{1/2 \leq t \leq 1} \frac{\log \mathbb{E}[b^{1-t} W^{2t}]}{t \log b}.$$

Putting $q = 2t$ shows that the expression being maximized is

$$1 - \frac{\log \mathbb{E}[b^{1-t} W^{2t}]}{t \log b} = 2a(q).$$

Thus (1) and (2) give

$$D_F = 2 \dim_{\text{Fr}} \mu_\infty. \quad (3)$$

In the squared-critical case, $H(2) = 0$, so the maximum is at $q = 2$, and the first branch of formula (1.4) yields the same identity:

$$D_F = 1 - \log_b \mathbb{E} W^2 = 2a(2) = 2 \dim_{\text{Fr}} \mu_\infty.$$

It remains to convert (3) into restriction sharpness. Lemma 1 below shows that $\mathcal{R}_r(\mu_\infty)$ implies

$$\dim_{\text{Fr}} \mu_\infty \geq \frac{2}{r'}.$$

Using (3),

$$\frac{2}{r'} \leq \frac{D_F}{2},$$

which is equivalent to

$$r \leq \frac{4}{4 - D_F}.$$

Therefore the restriction estimate fails for every larger r . Conversely, Corollary 1.6 of [1] proves it for every strictly smaller r . This proves the asserted value of the supremum. $\square$

Lemma 1 (Frequency localization). *Let μ be a finite Borel measure on $\mathbb{R}$. If $\mathcal{R}_r(\mu)$ holds for some $1 \leq r < 2$, then*

$$\mu(I) \lesssim |I|^{2/r'}$$

for every interval I of length at most 1. In particular, $\dim_{\text{Fr}} \mu \geq 2/r'$.

Proof. Choose a Schwartz function ϕ such that $\hat{\phi} \geq 1$ on $[-1, 1]$. If I has center ξ_0 and length $\delta \leq 1$, put

$$f_{\delta, \xi_0}(x) = \delta e^{2\pi i \xi_0 x} \phi(\delta x).$$

Then

$$\hat{f}_{\delta, \xi_0}(\xi) = \hat{\phi}\left(\frac{\xi - \xi_0}{\delta}\right), \quad \|f_{\delta, \xi_0}\|_r = \delta^{1-1/r} \|\phi\|_r = \delta^{1/r'} \|\phi\|_r.$$

Since $\hat{f}_{\delta, \xi_0} \geq 1$ on I , the assumed restriction estimate gives

$$\mu(I)^{1/2} \leq \|\hat{f}_{\delta, \xi_0}\|_{L^2(\mu)} \lesssim \delta^{1/r'}.$$

Squaring proves the claim. □

5 Explicit lognormal consequence

For

$$W = e^{\sigma N - \sigma^2/2}, \quad N \sim \mathcal{N}(0, 1),$$

the nondegenerate range is $0 < \sigma^2 < 2 \log b$, and the squared-critical/supercritical range is $\sigma^2 \geq \frac{1}{2} \log b$. In this range [1] gives

$$D_F = 2 \left(1 - \frac{\sigma}{\sqrt{2 \log b}} \right)^2.$$

Consequently

$$r_* = \frac{4}{4 - 2 \left(1 - \frac{\sigma}{\sqrt{2 \log b}} \right)^2}.$$

6 Verification notes and limitations

1. The localization lemma is deterministic and uses only Fourier scaling. It also checks the direction of the necessary inequality: a larger r forces a larger Frostman exponent.
2. The calculus step uses only convexity of $q \mapsto \log \mathbb{E} W^q$. The squared phase condition is exactly the sign condition $H(2) \leq 0$.
3. The identity $D_F = 2 \dim_{\text{Fr}} \mu_\infty$ is special to the squared-critical/supercritical phase. In the squared-subcritical phase the Frostman exponent can be strictly larger than $D_F/2$, so the same argument does not determine the optimal restriction exponent.
4. The proof identifies the supremum of valid exponents. It does not prove or disprove $\mathcal{R}_{r_*}(\mu_\infty)$ at the endpoint.
5. The exact Frostman spectrum formula (1) is an external standard cascade theorem. A reviewer should check its hypotheses against the stated all-moments and nonlattice assumptions; these are stronger than the usual hypotheses in the present regime.

7 Novelty check

The bounded search covered the run’s local 2000–2026 full-source index, exact-phrase searches for the source question, searches combining “Mandelbrot cascade”, “optimal exponent”, “restriction”, and “supercritical”, and citations of arXiv:2409.13164 through 23 July 2026. No paper stating Theorem 1 was found.

Carnovale–Fraser–de Orellana [4] give later L^q -dimension based sufficient estimates and explicitly improve the original Stein–Tomas range for some cascade parameters. They do not state the matching localization obstruction or determine the true critical exponent in the present phase. Thus the result here is a direct combination of known exact dimensions with a new sharpness identification, not a claimed new estimate in the squared-subcritical regime.

8 Human-review recommendation

Promote as a substantial partial solution if the reviewer confirms the standard exact Frostman formula (1) under the source hypotheses. The most important review points are the convention for the cascade L^q -spectrum, the equality case at squared criticality, and the interpretation of “optimal exponent” as the supremum when endpoint attainment is unknown.

References

- [1] X. Chen, Y. Han, Y. Qiu, and Z. Wang, *Harmonic analysis of Mandelbrot cascades – in the context of vector-valued martingales*, arXiv:2409.13164v3 (2025).
- [2] G. M. Molchan, *Scaling exponents and multifractal dimensions for independent random cascades*, Commun. Math. Phys. **179** (1996), 681–702.
- [3] J. Barral, A. Kupiainen, M. Nikula, E. Saksman, and C. Webb, *Critical Mandelbrot cascades*, Commun. Math. Phys. **325** (2014), 685–711; arXiv:1206.5444.
- [4] M. Carnovale, J. M. Fraser, and A. E. de Orellana, *Fourier restriction estimates based on L^q -dimensions: beyond Stein–Tomas*, arXiv:2606.07143 (2026).